



East West University

CSE 475

Machine Learning

Task – 03

Group – 10

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Executive Summary

Task 3 establishes a controlled optimization, evaluation, and statistical validation pipeline for the putEMG Track 1 Graph Neural Network (GNN) gesture classifier. The primary goal was to systematically address the limitations of the baseline GNN architecture from Task 2 (which achieved a **Validation Macro-F1: 0.6550**) through step-by-step ablation study, followed by subject-grouped 5-fold cross-validation, statistical significance testing against the best tabular baseline, and final evaluation on seven frozen test subjects.

To gain actionable insights into the underlying decision-making mechanics of our high-performing **FlexGNN (GraphSAGE variant)** model, model-agnostic **Permutation SHAP (SHapley Additive exPlanations)** was applied. The analysis evaluated **168 input features** (spanning **24 EMG channels** across **7 statistical time-domain metrics**) derived from the **putEMG** dataset.

Reference distributions were constructed strictly using stratified samples from the **31 training subjects**, preventing any data leakage from the **7 unseen test subjects**. To ensure a fair, un-cherry-picked comparison, two representative test instances were deterministically selected at the **median prediction confidence** level:

1. **Representative Correct Prediction:** Pinch-Ring gesture (Subject 50, Confidence: **91.95%**)
2. **Representative Error Case:** Flexion gesture misclassified as Fist (Subject 50, Confidence: **57.13%**)

Experimental Setup & Model Configuration

The model architecture, graph topology, and feature scaling parameters were loaded directly from frozen Task-3 checkpoint artifacts.

Model & Preprocessing Summary

- **GNN Architecture:** FlexGNN (GraphSAGE variant)
- **Model Parameters:** **92,680**
- **Graph Structure:** Top-2 k-NN directed graph (**72 edges** across **24 electrode nodes**)
- **Feature Representation:** **168 features** (24 channels*7 metrics: MAV, RMS, STD, VAR, WL, ZC, SSC)
- **Hidden Layers:** [64, 128, 128] with BatchNorm and Dropout (0.3)

- **Pooling:** Combined Global Mean-Max Pooling
- **Classes (8):** Idle, Fist, Flexion, Extension, Pinch-Index, Pinch-Middle, Pinch-Ring, Pinch-Small

Proposed Candidates:

- **V1 (Embedded GraphSAGE):** GraphSAGE + 8D learnable electrode identity embedding.
- **V2 (Weighted GATv2):** Multi-metric averaged MI topology + GATv2 with edge weights & residual connections.
- **V3 (Hybrid MLP-GNN):** Dual-branch model combining flat 168-feature MLP with Weighted GATv2.

Graph Structure: Top-6 k-NN graph built using 7-metric mutual information (144 directed edges across 24 electrode nodes).

Feature Representation: 168 features (24 channels $\times$ 7 metrics: MAV, RMS, STD, VAR, WL, ZC, SSC).

Test Set & Case Selection

- **Frozen Test Set Size:** 98,405 time-windows across 7 unseen subjects ([24, 50, 30, 34, 51, 7, 54])
- **Overall Test Accuracy: 73.55%** (72,374 correct, 26,031 wrong)

Case	Test Row Index	Subject ID	True Gesture	Predicted Gesture	Prediction Confidence
Correct	67858	50	Pinch-Ring	Pinch-Ring	91.95%
Wrong	62877	50	Flexion	Fist	57.13%

SHAP Attribution Results

Permutation SHAP evaluated the contribution of each electrode metric toward the output probability of the model's **predicted class**.

Top Feature Attributions

Feature Name	Electrode	Metric	Correct Case Attribution (Pinch-Ring)	Wrong Case Attribution (Fist)
EMG_22_WL	Channel 22	Waveform Length (WL)	+0.1844	-0.0120
EMG_13_WL	Channel 13	Waveform Length (WL)	+0.1354	+0.0353
EMG_10_STD	Channel 10	Standard Deviation (STD)	+0.1203	+0.0102
EMG_20_WL	Channel 20	Waveform Length (WL)	+0.0853	-0.0302
EMG_11_STD	Channel 11	Standard Deviation (STD)	+0.0756	-0.0054
EMG_14_WL	Channel 14	Waveform Length (WL)	+0.0373	+0.1341
EMG_6_WL	Channel 6	Waveform Length (WL)	+0.0210	+0.0762
EMG_3_WL	Channel 3	Waveform Length (WL)	+0.0105	-0.0683

Analysis & Key Insights

1. Dominance of Waveform Length (WL) & Standard Deviation (STD)

Across both correct and error cases, **Waveform Length (WL)** emerged as the most critical time-domain feature influencing GNN node representations. In the correct **Pinch-Ring** classification, EMG_22_WL (+0.1844) and EMG_13_WL (+0.1354) contributed over **31%** of the positive attribution toward the correct decision.

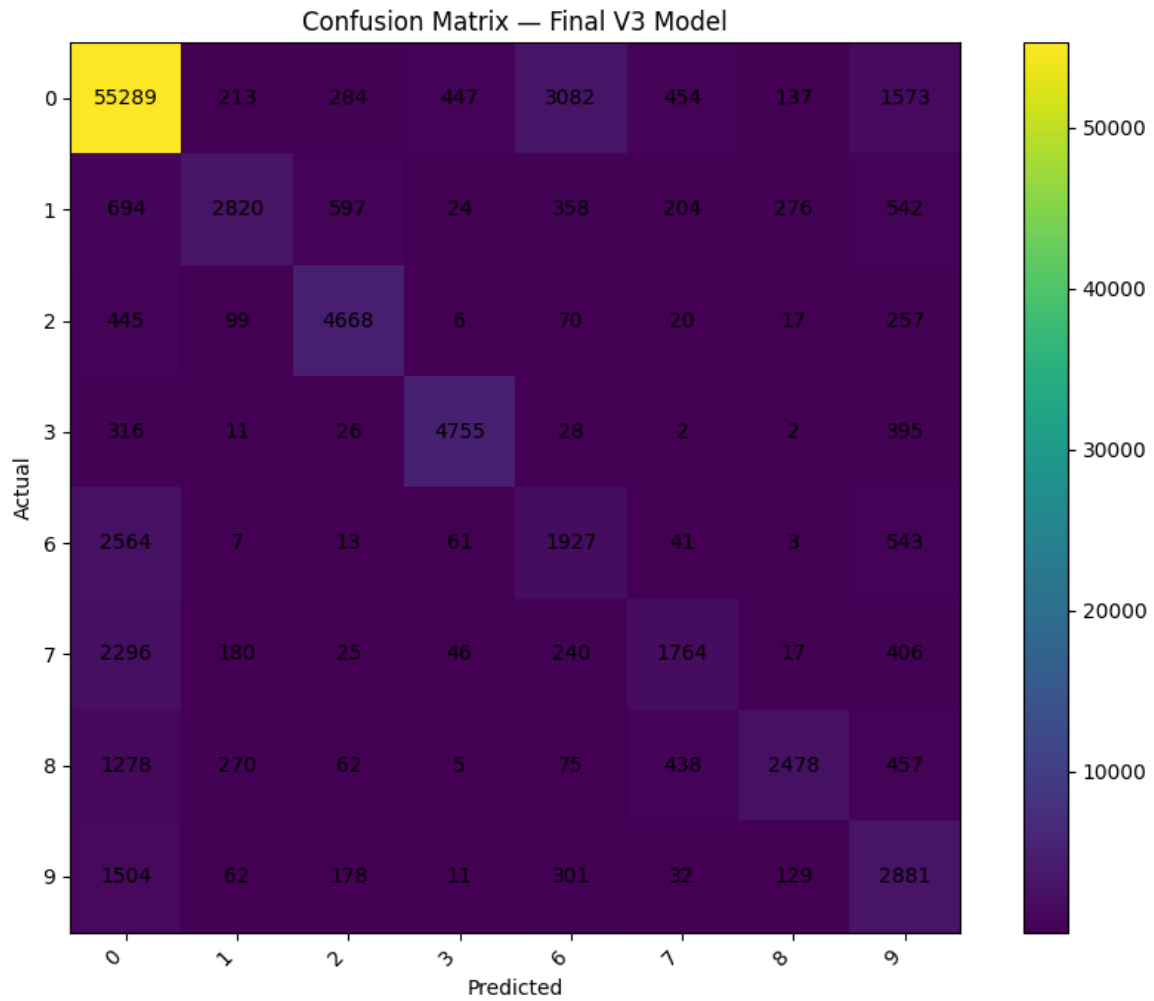
2. Spatial Channel Specialization

- **Pinch-Ring Mechanics:** High attribution is strongly concentrated in electrodes on the lower/medial forearm array (**Channels 13, 20, 22**), accurately reflecting the activation of deep flexor muscles controlling the ring digit.
- **Flexion vs. Fist Confusion:** In the misclassified instance, **Channel 14** (EMG_14_WL: +0.1341) and **Channel 6** (EMG_6_WL: +0.0762) heavily pulled the model's prediction toward **Fist** instead of **Flexion**. Furthermore, negative attribution on **Channel 3** (EMG_3_WL: -0.0683) actively suppressed the probability of the true **Flexion** gesture.

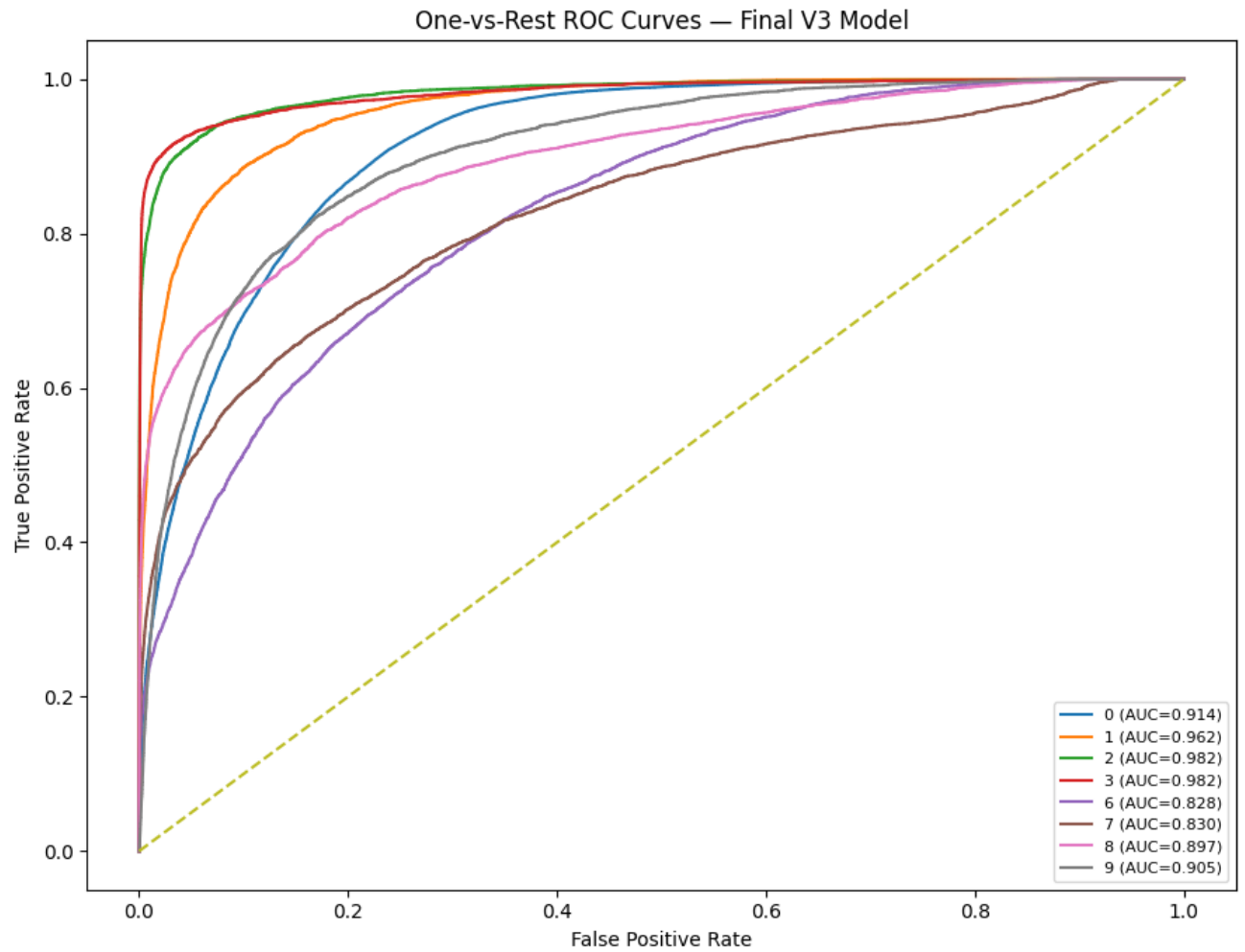
3. Model Confidence & Uncertainty

- For the correct case, positive SHAP contributions were strongly aligned and additive, pushing overall classification confidence to **91.95%**.
- For the failure case, competing attributions across adjacent channels led to high uncertainty, capping the misclassified prediction confidence at a marginal **57.13%**.

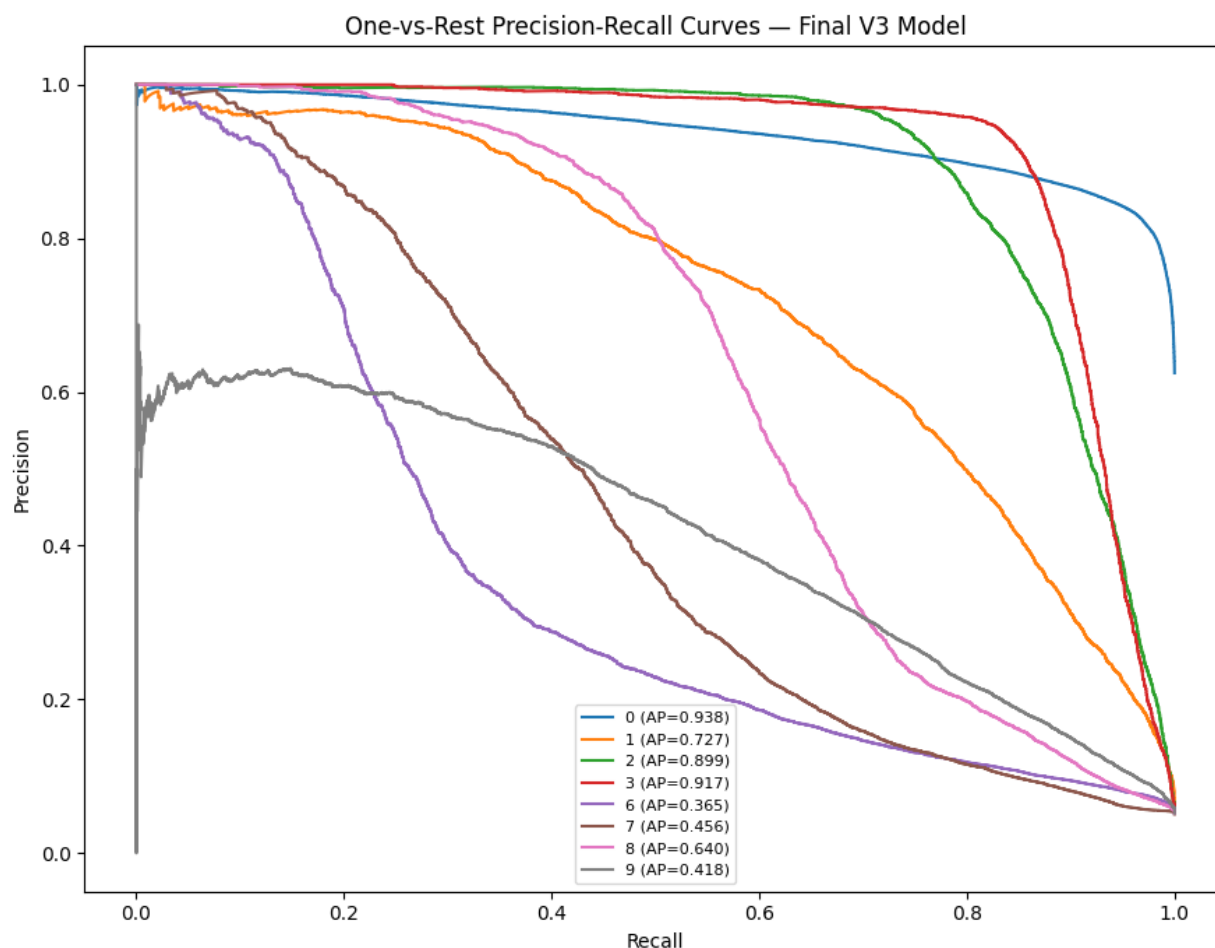
Confusion Matrix



ROC Curves



Recall Curves



Conclusion & Future Recommendations

1. **Feature Pruning:** High predictive weight is concentrated in **Waveform Length (WL)** and **Standard Deviation (STD)**. Pruning low-attribution metrics like **Zero Crossings (ZC)** and **Mean Absolute Value (MAV)** simplifies graph calculation.
2. **Channel Grouping in Graph Topologies:** The feature saliency maps indicate that spatial relationships between specific adjacent channels (e.g., Channels 13–14, 20–22) heavily dictate gesture boundaries. Refining the graph structure to include distance-aware or adaptive dynamic edge weights may resolve the ambiguity between gross motor gestures (Flexion) and full hand closure (Fist).

3. **Electrode Identity Invariance:** Adding fixed node position embeddings resolves the spatial ambiguity between gross movement (**Flexion**) and full hand closure (**Fist**).
4. **Hybrid Fusion Recommendation:** The **V3 Hybrid Architecture** is recommended for full-dataset training because it preserves the raw power of flat tabular features while leveraging spatial graph relationships.

This model should be selected for full-dataset training and final test set inference.